3

Theory

3.1 INTRODUCTION

In Chapter 1 we introduced the fundamental concepts concerning a new type of surface wave that can be excited only on finite periodic structures. It was pointed out that radiation could occur from such surface waves and therefore could lead to an increase in the RCS level in the backward direction. Similarly, if the structure was active—as, for example, for a phased array—this type of surface wave could lead to a very significant variation of the terminal impedance form element to element. This could make precise matching difficult, if not impossible.

Furthermore, in Chapter 2 we presented the theory for the RCS of antennas in general. In particular, it was shown that the scattering from antennas can be decomposed into two components, namely the antenna mode component being proportional to the square of the reflection coefficient observed at the antenna terminals and another called for the residual component (also earlier denoted structural). Although the precise definition of the second was somewhat illusive, we nevertheless demonstrated that it was equal to zero for large flat apertures with uniform illumination. The most prominent member of that family was without a doubt the antenna array. In fact it has a very unique position in the world of radiators.

So far our mode of attack has been mostly based on physical insight. That pointed us in the right direction without going through an enormous amount of tedious calculations that might have led us into dead ends.

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In my first book [61] we investigated primarily structures of infinite × infinite extent. However, in this chapter we shall primarily develop the mathematical tools that will enable us to investigate the finite × infinite case.

More specifically, we shall investigate arrays comprised of infinitely long columns along the z axis, where the elements have arbitrary orientation $\hat{p} = \hat{x}p_x + \hat{y}p_y + \hat{z}p_z$. Such single-column arrays are often called stick arrays. They are simply the building blocks for more complicated arrays and are therefore extremely important to investigate.

3.2 THE VECTOR POTENTIAL AND THE *H* FIELD FOR COLUMN ARRAYS OF HERTZIAN ELEMENTS

Consider an array of Hertzian elements with length dl and arbitrary orientation $\hat{p}$ as shown in Fig. 3.1. The reference element is located at (0, 0, z') while the element current are assumed to be

$$I_m = I_0\, e^{-j\beta m D_z s_z}, \tag{3.1}$$

where m denotes the element number along the z axis and I_0 denotes the current in reference element $m = 0$.

In fact, when the reference element is located at the origin (i.e., $z' = 0$), we have already determined the vector potential $d\overline{A}_q$ for this case in reference 62 as

$$d\overline{A}_q = \hat{p}\frac{\mu I_0\, dl}{4jD_z}\sum_{n=-\infty}^{\infty} e^{-j\beta z r_z} H_0^{(2)}(\beta r_\rho \rho), \tag{3.2}$$

where

$$r_z = s_z + n\frac{\lambda}{D_z}, \tag{3.3}$$

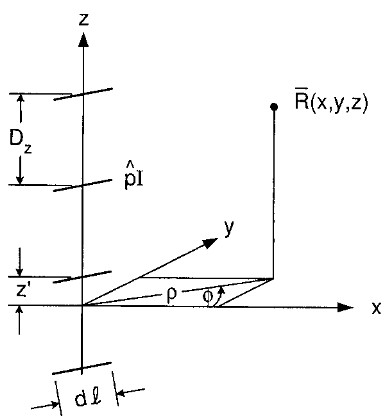

Fig. 3.1 Infinite stick array of Hertzian dipoles with arbitrary orientation $\hat{p} = \hat{x}p_x + \hat{y}p_y + \hat{z}p_z$ and element currents $I_m = I_0\, e^{-j\beta m D_z r_z}$. The reference element is located at (0, 0, z').

$$r_\rho = \sqrt{1 - r_z^2} = \sqrt{1 - \left(s_z + n\frac{\lambda}{D_z}\right)^2}. \tag{3.4}$$

We shall next generalize to the case where the reference element 0 is located at (0, 0, z') as shown in Fig. 3.1. This situation corresponds to keeping the reference element at the origin and lowering $\overline{R}$ to the new position (0, 0, $z - z'$). According to (3.2), we then find the vector potential for the general case to be

$$d\overline{A}_q = \hat{p}\frac{\mu I_0\, dl}{4jD_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} H_0^{(2)}(\beta r_\rho \rho). \tag{3.5}$$

for reference element at (0, 0, z').

Next we obtain the H field at the point of observation R from

$$d\overline{H} = \frac{1}{\mu}\nabla \times d\overline{A}. \tag{3.6}$$

Substituting (3.5) into (3.6), we obtain

$$d\overline{H}_q = \frac{I_0\, dl}{4jD_z}\sum_{n=-\infty}^{\infty} \nabla \times [\hat{p}\, e^{-j\beta(z-z')r_z} H_0^{(2)}(\beta r_\rho \rho)].$$

Further applying the vector identity

$$\nabla \times (\hat{p}\phi) = \phi\nabla \times \hat{p} - \hat{p} \times \nabla\phi$$

(where we note that the first term on the right-hand side is zero because $\hat{p}$ is a constant), we obtain

$$d\overline{H}_q = -\frac{I_0\, dl}{4jD_z}\sum_{n=-\infty}^{\infty} \hat{p} \times \nabla[e^{-j\beta(z-z')r_z} H_0^{(2)}(\beta r_\rho \rho)]. \tag{3.7}$$

In *cylindrical coordinates* we obtain from (3.7)

$$\begin{aligned} d\overline{H}_q &= -\frac{I_0\, dl}{4jD_z}\sum_{n=-\infty}^{\infty} \hat{p} \\ &\quad \times \left[\hat{\rho}\frac{\partial}{\partial\rho} + \hat{\phi}\frac{1}{\rho}\frac{\partial}{\partial\phi} + \hat{z}\frac{\partial}{\partial z}\right][e^{-j\beta(z-z')r_z} H_0^{(2)}(\beta r_\rho \rho)] \\ d\overline{H}_q &= -\frac{\beta I_0\, dl}{4jD_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} \\ &\quad \cdot [\hat{p} \times \hat{\rho} r_\rho H_0^{(2)'}(\beta r_\rho \rho) - j\hat{p} \times \hat{z} r_z H_0^{(2)}(\beta r_\rho \rho)]. \end{aligned} \tag{3.8}$$

In *rectangular coordinates* we obtain from (3.7)

$$d\overline{H}_q = -\frac{I_0\,dl}{4jD_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}$$
$$\cdot\ \hat{p}\times\left[\hat{x}\frac{\partial}{\partial x}H_0^{(2)}(\beta r_\rho\sqrt{x^2+y^2}) + \hat{y}\frac{\partial}{\partial y}H_0^{(2)}(\beta r_\rho\sqrt{x^2+y^2})\right.$$
$$\left. -\ j\hat{z}\beta r_z H_0^{(2)}(\beta r_\rho\sqrt{x^2+y^2})\right]. \tag{3.9}$$

Equations (3.8) and (3.9) are valid for arbitrary $\hat{p}$. However, in the following investigation we shall specialize to:

I: Longitudinal case where $\hat{p}$ is parallel with the array axis; that is, in our case $\hat{p} = \hat{z}$.

II: Transverse case where $\hat{p}$ is orthogonal to the array axis. In our case we will further specialize to $\hat{p} = \hat{x}$.[1]

3.3 CASE I: LONGITUDINAL ELEMENTS

For $\hat{p} = \hat{z}$ we obtain from (3.8) for Hertzian elements

$$d\overline{H}_q = -\frac{\beta I_0\,dl}{4jD_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}\hat{\phi} r_\rho H_0^{(2)'}(\beta r_\rho \rho). \tag{3.10}$$

Furthermore,

$$\begin{aligned}\overline{E}_q &= \frac{1}{j\omega\varepsilon}\nabla\times\overline{H}_q \\ &= \frac{1}{j\omega\varepsilon}\left[\hat{\rho}\left[\frac{1}{\rho}\frac{\partial H_z}{\partial\phi} - \frac{\partial H_\phi}{\partial z}\right] + \hat{\phi}\left[\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial\rho}\right]\right. \\ &\quad \left. +\ \hat{z}\left[\frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho H_\phi) - \frac{1}{\rho}\frac{\partial H_\rho}{\partial\phi}\right]\right].\end{aligned} \tag{3.11}$$

Substituting (3.10) into (3.11), we obtain

$$d\overline{E}_q = \frac{-1}{j\omega\varepsilon}\frac{\beta I_0\,dl}{4jD_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}$$
$$\cdot\left[j\hat{\rho} r_\rho r_z \beta H_0^{(2)'}(\beta r_\rho\rho) + \hat{z} r_\rho^2\beta\left[\frac{1}{\rho}\rho H_0^{(2)''}(\beta r_\rho\rho) + \frac{1}{\beta r_\rho\rho}H_0^{(2)'}(\beta r_\rho\rho)\right]\right]. \tag{3.12}$$

[1] This choice in no way limits the generality of our investigation since our point of observation $\overline{R}(x, y, z)$ is arbitrary.

Applying $\beta = \omega\sqrt{\mu\varepsilon}$, $Z = \sqrt{\mu/\varepsilon}$, and the recursive relationship

$$H_0^{(2)''}(\beta r_\rho \rho) + \frac{1}{\beta r_\rho \rho} H_0^{(2)'}(\beta r_\rho \rho) = -H_0^{(2)}(\beta r_\rho \rho),$$

we can write (3.12) as

$$d\overline{E}_q = \frac{\beta Z I_0\, dl}{4D_z} \sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} \cdot [-j\hat{\rho} r_\rho r_z H_0^{(2)'}(\beta r_\rho \rho) + \hat{z} r_\rho^2 H_0^{(2)}(\beta r_\rho \rho)], \tag{3.13}$$

where $\hat{p} = \hat{z}$ and the reference element is located at $(0, 0, z')$.

3.3.1 Total Field from Infinite Column Array of *z*-Directed Elements of Arbitrary Length 2*l*

Equation (3.13) yields the entire field from an infinite array of z-directed (collinear) elements of length dl and the constant current I_0. In other words, the expression is essentially a dyadic Green's function.

Thus, we can use this expression to obtain the total field $\overline{E}_q$ of an infinite collinear array of elements of arbitrary length as well as arbitrary currents $I(z')$ simply by integrating $d\overline{E}^{(q)}$ given by (3.13) over a single element as discussed in reference 63.

More specifically, let the endpoints of the reference element be denoted by the points "a" and "b" as illustrated in Fig. 3.2. Further noting that $H_0^{(2)'}(\beta r_\rho \rho) =$

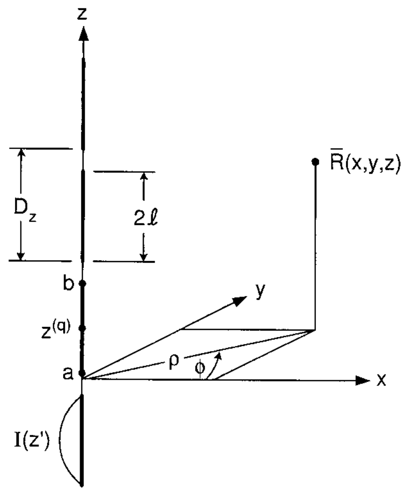

Fig. 3.2 Infinite collinear array of z-directed elements of length 2l and current distribution $I(z')$. The reference element is located at $(0, 0, z^{(q)})$.

$-H_1^{(2)}(\beta r_\rho \rho)$, we obtain the total field $\overline{E}^{(q)}$

$$\overline{E}^q = -\frac{\beta Z}{4D_z} \sum_{n=-\infty}^{\infty} \int_{z'=a}^{b} I(z')\, e^{-j\beta(z-z')r_z}\, dz' \cdot [-j\hat{\rho} r_\rho r_z H_1^{(2)}(\beta r_\rho \rho) + \hat{z} r_\rho^2 H_0^{(2)}(\beta r_\rho \rho)]. \tag{3.14}$$

The integral in (3.14) can be better managed if we express the source point z' with respect to a reference point $z^{(q)}$ that basically can be chosen anywhere on the reference element. It is natural to choose it as the midpoint of an element of length $2l$, that is

$$z' = z^{(q)} + z'', \qquad -l < z'' < l. \tag{3.15}$$

Substituting (3.15) into (3.14), we obtain

$$\overline{E}^q = -\frac{\beta Z I^{(q)}}{4D_z} \sum_{n=-\infty}^{\infty} e^{-j\beta(z-z^{(q)})r_z} P_z^{(q)} \cdot [j\hat{\rho} r_\rho r_z H_1^{(2)}(\beta r_\rho \rho) + \hat{z} r_\rho^2 H_0^{(2)}(\beta r_\rho \rho)], \tag{3.16}$$

where the pattern factor $P_z^{(q)}$ normalized to the reference current $I^{(q)}(z)$ at the reference point $z^{(q)}$ is defined as

$$P_z^{(q)} = \frac{1}{I^{(q)}} \int_{-l}^{l} I^{(q)}(z'')\, e^{j\beta z'' r_z}\, dz.'' \tag{3.17}$$

Equation (3.16) gives us the total E field of an infinite collinear column array of z-directed elements of length $2l$, the arbitrary current distribution $I^{(q)}(z)$, and the reference point of the reference element located at $z^{(q)}$.

3.3.2 The Voltage Induced in an Element by an External Field

Ultimately we are going to determine the mutual impedance $Z^{q,'q}$ between a collinear array q and the reference element of another array q'. It will be recalled that the mutual impedance in general is merely defined as the negative of the voltage induced in the external element q' by the entire line array divided by the current on the reference element of array $\underline{q}$. The voltage induced at the terminals of a single element by an arbitrary field $\overline{E}$ is given by [64]

$$V^{(q')} = \frac{1}{I^{q't}} \int \overline{E} \cdot \hat{p}^{(q')} I^{q't}(l)\, dl, \tag{3.18}$$

where $\hat{p}^{(q')}$ is the direction of element in question, $I^{q't}(l)$ is the current distribution of the element q' when transmitting from its terminals, and $I^{q't}(0)$ is the current at the terminals of q'.

In the present case of interest we are considering two collinear arrays in which case

$$\hat{p}^{q'} = \hat{z} \quad \text{and} \quad \rho \rightarrow \rho'. \tag{3.19}$$

Substituting (3.16) and (3.19) into (3.18) yields

$$V^{(q')} = -\frac{\beta Z I^{(q)}}{4D_z} \sum_{n=-\infty}^{\infty} P_z^{(q)} \frac{1}{I^{q't}} \int_{\text{Element } q'} I^{q't}\, e^{-j\beta(z-z^{(q)})r_z}\, dz\, r_\rho^2 H_0^{(2)}(\beta r_\rho \rho)'. \tag{3.20}$$

Similar to earlier [see (3.15)], we now express the arbitrary point z on the external element q' with respect to a reference point $z^{(q')}$:

$$z = z^{(q')} + z'', \qquad -l' < z'' < l'. \tag{3.21}$$

Substituting (3.21) into (3.20), we obtain

$$V^{(q')} = -\frac{\beta Z I^{(q)}}{4D_z} \sum_{n=-\infty}^{\infty} e^{-j\beta(z^{(q')}-z^{(q)})r_z} P_z^{(q)} P^{q't} r_\rho^2 H_0^{(2)}(\beta r_\rho \rho'), \tag{3.22}$$

where the normalized transmitting pattern function for the external element q' is given by

$$P^{q't} = \frac{1}{I^{q't}(z^{q'})} \int_{-l'}^{l} I^{q't}(z'')\, e^{-j\beta z'' r_z}\, dz''. \tag{3.23}$$

3.3.3 The Mutual Impedance $Z^{q,'q}$ Between a Column Array q and an External Element q'

As indicated above, the mutual impedance between a column array q and a single external element (namely the reference element of another array q') is simply defined as

$$Z^{q',q} = -\frac{V^{(q')}}{I^{(q)}}. \tag{3.24}$$

Substituting (3.22) into (3.24) yields

$$Z^{q',q} = \frac{\beta Z}{4D_z} \sum_{n=-\infty}^{\infty} e^{-j\beta(q'-q)r_z} r_\rho^2 P_z^{(q)} P_z^{q't} H_0^{(2)}(\beta r_\rho \rho'), \tag{3.25}$$

where the two pattern functions $P_z^{(q)}$ and $P_z^{q't}$ are defined by (3.17) and (3.23), respectively.

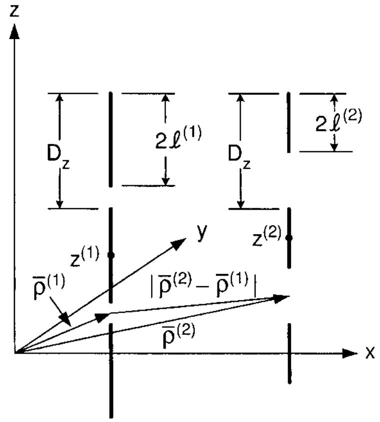

Fig. 3.3 Two infinite arrays of z-directed elements with length $2l^{(1)}$ and $2l^{(2)}$ and the reference points at $(\overline{\rho}^{(1)}, Z^{(1)})$ and $(\overline{\rho}^{(2)}, Z^{(2)})$, respectively.

In the case as shown in Fig. 3.3, it is easy to generalize (3.25) to

$$Z^{q',q} = \frac{\beta Z}{4D_z} \sum_{n=-\infty}^{\infty} e^{-j\beta(q'-q)r_z} r_\rho^2 P_z^{(q)} P_z^{q't} H_0^{(2)}(\beta r_\rho |\overline{\rho}' - \overline{\rho}|) \tag{3.26}$$

where the two pattern functions are unchanged.

Let us finally consider the more general case where the reference points of the column arrays are given by $(\overline{\rho}^{(1)}, Z^{(1)})$ and $(\overline{\rho}^{(2)}, Z^{(2)})$, respectively, as shown in Fig. 3.3. By generalization of (3.25) we then readily obtain

$$Z^{2,1} = \frac{\beta Z}{4D_z} \sum_{n=-\infty}^{\infty} e^{-j\beta(Z^{(2)}-Z^{(1)})r_z} r_\rho^2 P_z^{(2)t} P_z^{(1)} H_0^{(2)}(\beta r_\rho |\overline{\rho}^{(2)} - \overline{\rho}^{(1)}|), \tag{3.27}$$

where the two pattern functions are unchanged as defined by (3.17) and (3.23).

Equation (3.26) for the mutual impedance $Z^{2,1}$ between a *column* array at $\overline{\rho}^{(1)}, Z^{(1)}$ and the reference element of another array at $\overline{\rho}^{(2)}, Z^{(2)}$ should be compared to an earlier result for the mutual impedance $Z^{2,1}$ between a *planar* array and the reference element of another array found to be [65]

$$Z^{2,1} = \frac{Z}{2D_x D_z} \sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{e^{-j\beta(\overline{R}^{(2)} - \overline{R}^{(1)})\cdot\hat{r}}}{r_y} [{}_\perp P^{(2)t} {}_\perp P^{(1)} + {}_\parallel P^{(2)t} {}_\parallel P^{(1)}], \tag{3.28}$$

where

$$\begin{aligned} {}^{\perp}_{\parallel} P^{(1)} &= \hat{p}^{(1)} \cdot {}^{\perp}_{\parallel}\hat{n} P^{(1)}, \\ {}^{\perp}_{\parallel} P^{(2)t} &= \hat{p}^{(2)} \cdot {}^{\perp}_{\parallel}\hat{n} P^{(2)t}, \end{aligned}$$

If we limit ourselves to scan the yz plane, the orthogonal terms $_{\perp}P^{(1)}$ are either zero or cancel when k varies. We then note that (3.27) and (3.28) are similar as far as variation in the z direction is concerned, which should not surprise us since Floquet's theorem is valid for both cases in this direction. However, their behavior in the xy plane clearly indicates a planar wave in the double infinite case and a cylindrical wave in the collinear case.

Note further that for $\hat{p}^{(1)} = \hat{p}^{(2)} = \hat{z}$ we have

$$\hat{p}^{(1)} \cdot {}_{\perp}\hat{n} = \hat{p}^{(2)} \cdot {}_{\perp}\hat{n} = 0,$$
$$\hat{p}^{(1)} \cdot {}_{\parallel}\hat{n} = \hat{p}^{(2)} \cdot {}_{\parallel}\hat{n} = -r_\rho.$$

3.4 CASE II: TRANSVERSE ELEMENTS

We shall next investigate the transverse case, in particular where the line array is oriented in the x direction and the reference element located at $\overline{R}'(0, 0, z')$ while the point of observation is $\overline{R}(x, y, z)$, as shown in Fig. 3.4.

From (3.9) we obtain for $\hat{p} = \hat{x}$ and $dl \to dx'$

$$d\overline{H}_q = -\frac{I_0\, dx'}{4jD_z} \sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} \left[\hat{z}\frac{\partial}{\partial y} H_0^{(2)}(\beta r_\rho \sqrt{x^2+y^2}) \right.$$
$$\left. + j\hat{y}\beta r_z H_0^{(2)}(\beta r_\rho \sqrt{x^2+y^2})\right]. \qquad (3.29)$$

Furthermore, we have in general

$$\overline{E}_q = \frac{1}{j\omega\varepsilon} \nabla \times \overline{H}_q$$

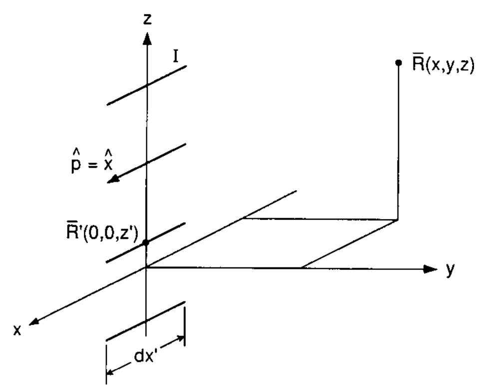

Fig. 3.4 An infinite array of transverse Hertzian dipoles with orientation $\hat{p} = \hat{x}$. The reference element is located at $\overline{R}'(0, 0, z')$ while the point of observation is at $\overline{R}(x, y, z)$.

$$= \frac{1}{j\omega\varepsilon}\left[\hat{x}\left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right) + \hat{y}\left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x}\right) + \hat{z}\left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}\right)\right]. \tag{3.30}$$

Substituting (3.29) into (3.30):

$$\begin{aligned} d\overline{E}_q = & -\frac{1}{j\omega\varepsilon}\frac{I_0\,dx'}{4jD_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} \\ & \cdot\left[\hat{x}\left[\frac{\partial^2}{\partial y^2}H_0^{(2)}(\beta r_\rho\sqrt{x^2+y^2}) - \beta^2 r_z^2 H_0^{(2)}(\beta r_\rho\sqrt{x^2+y^2})\right]\right. \\ & \left. -\ \hat{y}\frac{\partial^2}{\partial x\partial y}H_0^{(2)}(\beta r_\rho\sqrt{x^2+y^2}) + j\hat{z}\beta r_z\frac{\partial}{\partial x}H_0^{(2)}(\beta r_\rho\sqrt{x^2+y^2})\right]. \end{aligned} \tag{3.31}$$

We are next going to investigate the *x, y*, and *z* components separately.

3.4.1 The *x* Component of $\overline{E}_q$

In order to reduce the x component of (3.31), consider for a moment the ∇^2 operator in cylindrical coordinates:

$$\nabla^2 = \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2}{\partial\phi^2} + \frac{\partial^2}{\partial z^2}.$$

Operating on $H_0^{(2)}(\beta r_\rho\rho)$ we obtain

$$\begin{aligned} \nabla^2 H_0^{(2)}(\beta r_\rho\rho) &= \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial H_0^{(2)}(\beta r_\rho\rho)}{\partial\rho}\right) \\ &= (\beta r_\rho)^2\left[H_0^{(2)''}(\beta r_\rho\rho) + \frac{1}{\beta r_\rho\rho}H_0^{(2)'}(\beta r_\rho)\right]. \end{aligned}$$

Furthermore, by applying the recursive formula

$$H_0^{(2)''} + \frac{1}{x}H_0^{(2)'} = -H_0^{(2)}$$

we obtain

$$\nabla^2 H_0^{(2)}(\beta r_\rho\rho) = -(\beta r_\rho)^2 H_0^{(2)}(\beta r_\rho\rho). \tag{3.32}$$

Expressed in rectangular coordinates we have

$$\nabla^2 H_0^{(2)}(\beta r_\rho\rho) = \frac{\partial^2 H_0^{(2)}(\beta r_\rho\rho)}{\partial x^2} + \frac{\partial^2 H_0^{(2)}(\beta r_\rho\rho)}{\partial y^2}. \tag{3.33}$$

Equating (3.32) and (3.33), we obtain

$$\frac{\partial^2 H_0^{(2)}(\beta r_\rho \rho)}{\partial y^2} = -\frac{\partial^2 H_0^{(2)}(\beta r_\rho \rho)}{\partial x^2} - (\beta r_\rho)^2 H_0^{(2)}(\beta r_\rho \rho). \tag{3.34}$$

Substituting (3.34) into (3.31) and assuming current distribution $I(x')$ on all the elements, we obtain the total x component of the E field by integrating over the reference element:

$$\begin{aligned}\overline{E}_q^{x,x'} &= -\hat{x}\frac{1}{\omega\varepsilon}\frac{1}{4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}\\ &\quad\times \int_{\text{Ref. ele.}} I(x')\left[\frac{\partial^2}{\partial x^2} + (\beta r_\rho)^2 + (\beta r_z)^2\right] H_0^{(2)}(\beta r_\rho \rho)\, dx',\end{aligned}$$

where ρ now is

$$\rho = \sqrt{(x-x')^2 + y^2}.$$

Further noting that $r_\rho^2 + r_z^2 = 1$ and $\dfrac{\partial^2}{\partial x'^2} = \dfrac{\partial^2}{\partial x^2}$, we obtain

$$\begin{aligned}\overline{E}_q^{x,x'} &= -\hat{x}\frac{\sqrt{\mu/\varepsilon}}{\omega\sqrt{\varepsilon\mu}4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}\\ &\quad\times \int_{\text{Ref. ele.}} I(x')\left[\frac{\partial^2}{\partial x'^2} + \beta^2\right] H_0^{(2)}(\beta r_\rho \rho)\, dx'.\end{aligned} \tag{3.35}$$

Integrating the first term in the integral of (3.35) twice by parts, we obtain

$$\begin{aligned}\overline{E}_q^{x,x'} &= -\hat{x}\frac{Z}{\beta 4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}\left[I(x')\frac{\partial}{\partial x'}H_0^{(2)}(\beta r_\rho \rho)\Big|_{\text{Ref. ele. endpts.}}\right.\\ &\quad\left. - \int_{\text{Ref. ele.}} I'(x')\frac{\partial}{\partial x'}H_0^{(2)}(\beta r_\rho \rho) - \beta^2 I(x') H_0^{(2)}(\beta r_\rho \rho)\, dx'\right]\\ &= -\hat{x}\frac{Z}{\beta 4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}\left[\left[I(x')\frac{\partial}{\partial x'}H_0^{(2)}(\beta r_\rho \rho)\right.\right.\\ &\quad\left. - I'(x')\ H_0^{(2)}(\beta r_\rho \rho)\right]\Big|_{\text{Ref. ele. endpts.}}\\ &\quad\left. + \int_{\text{Ref. ele.}} [I''(x') + \beta^2 I(x')]\right] H_0^{(2)}(\beta r_\rho \rho)\, dx.\end{aligned} \tag{3.36}$$

If the current distribution $I(x')$ is sinusoidal with propagation constant β, we note that the integrand in (3.36) equals zero. Furthermore, the current $I(x')$ at the endpoints of the reference element must equal zero (boundary condition). Thus, the first term in (3.36) is also zero and we finally have

$$\overline{E}_q^{x,x'} = \hat{x}\frac{Z}{\beta 4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}[I'(x')H_0^{(2)}(\beta r_\rho \rho)]_{\text{Ref. ele. endpts.}} \tag{3.37}$$

for sinusoidal current with propagation constant β. Specifically for a sinusoidal current distribution

$$I(x') = I_a \sin\beta(l' - |x'|) \qquad \text{for} - l' < x' < l' \tag{3.38}$$

comprised of two sections with endpoint at $x' = \pm l'$ and 0. We obtain from (3.37)

$$\begin{aligned}\overline{E}_q^{x,x'} &= -\hat{x}\frac{ZI_a}{4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}[H_0^{(2)}(\beta r_\rho \rho_+) + H_0^{(2)}(\beta r_\rho \rho_-)\\ &\quad - 2\cos\beta l' H_0^{(2)}(\beta r_\rho \rho_0)],\end{aligned} \tag{3.39}$$

where $\rho_\pm = \sqrt{(x \pm l')^2 + y^2}$ and $\rho_0 = \sqrt{x^2 + y^2}$ as illustrated in Fig. 3.5.

The voltage $V_q^{x,x'}$ induced at the terminals of an x-directed element with current distribution $I_x(x)$ is given by (see Fig. 3.6, top)

$$V_q^{x,x'} = \frac{1}{I_x(0)}\int_{\text{Ext. ele.}} I_x(x)\,\overline{E}_q^{x,x'}\,dx. \tag{3.40}$$

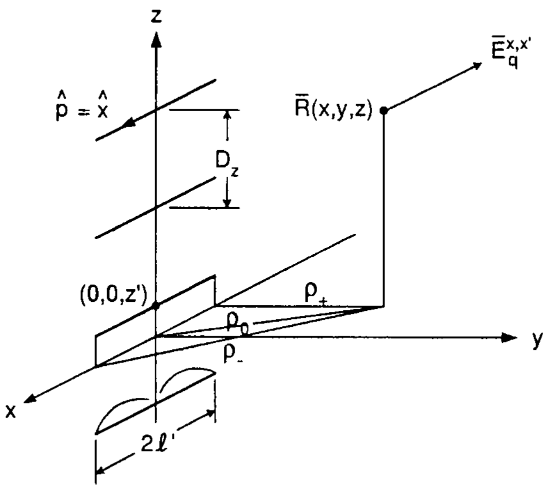

Fig. 3.5 The x component of the E field at point $\overline{R}(x, y, z)$ from a transverse array with element orientation $\hat{p} = \hat{x}$ and sinusoidal current distribution can be expressed by an infinite series of three Hankel functions with radii ρ_+, ρ_- and ρ_0 as shown in the figure. See (3.39) for details.

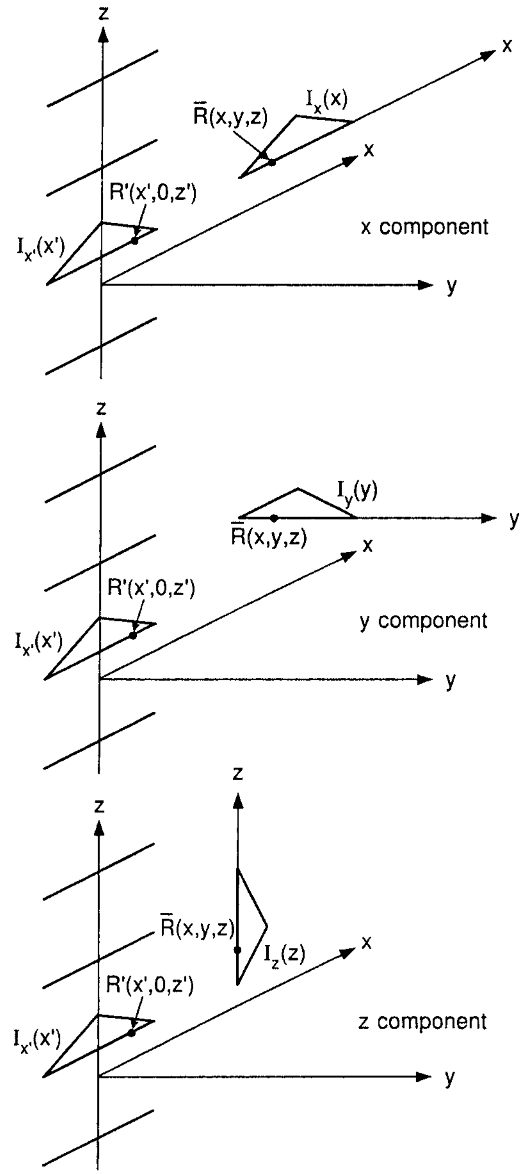

Fig. 3.6 An infinite line array with transverse x-directed elements of length 2l with current distribution $I_{x'}(x')$ induces a voltage in a single dipole with current distribution oriented along the x axis with $I_x(x)$ (top), y axis with $I_y(y)$ (middle), and z axis with $I_z(z)$ (bottom).

Finally, the mutual impedance $Z_q^{x,x'}$ between a stick array with an external element is defined as

$$Z_q^{x,x'} = -\frac{V_q^{x,x'}}{I(0)}, \tag{3.41}$$

where from (3.38)

$$I(0) = I_a \sin \beta l'. \tag{3.42}$$

Substituting (3.39), (3.40), and (3.42) into (3.41), we obtain

$$Z_q^{x,x'} = \frac{Z}{\beta 4D_z}\frac{1}{\sin \beta l'}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}\frac{1}{I_x(0)}\int_{\text{Ext. ele.}} I_x(x)$$
$$\cdot\, [H_0^{(2)}(\beta r_\rho \rho_+) + H_0^{(2)}(\beta r_\rho \rho_-) - 2\cos \beta l' H_0^{(2)}(\beta r_\rho \rho_0)]\, dx. \tag{3.43}$$

For further discussion of (3.43), see Section 3.5.

3.4.2 The y Component of $\overline{E}_q$

The y component $\overline{E}_q^{y,x'}$ of the total field $\overline{E}_q$ from a line array with current distribution $I_{x'}(x')$ is readily obtained from (3.31) by integration of the y component over the reference element with current distribution $I_{x'}(x')$:

$$\overline{E}^{y,x'} = \hat{y}\frac{\sqrt{\mu/\varepsilon}}{\omega\sqrt{\varepsilon\mu}4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}$$
$$\cdot\, \frac{\partial}{\partial y}\int_{\text{Ref. ele.}} I_{x'}(x')\frac{\partial}{\partial x'}H_0^{(2)}(\beta r_\rho\sqrt{(x-x')^2+y^2})\, dx', \tag{3.44}$$

where we have used $\dfrac{\partial}{\partial x} = -\dfrac{\partial}{\partial x'}$. Integration by parts once with respect to x' gives

$$\overline{E}_q^{y,x'} = \hat{y}\frac{Z}{\beta 4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}$$
$$\cdot\, \frac{\partial}{\partial y}\Big[I_{x'}(x')H_0^{(2)}(\beta r_\rho\sqrt{(x-x')^2+y^2})\, dx' \,|_{\text{Ref. ele. endpts.}}$$
$$- \int_{\text{Ref. ele.}} *I'_{x'}(x')H_0^{(2)}(\beta r_\rho\sqrt{(x-x')^2+y^2})\, dx\Big].' \tag{3.45}$$

Since $I_{x'}(x') = 0$ at the endpoints of the reference element, the first term in (3.45) must equal zero. Thus, the total y component at $\overline{R}(x, y, z)$ of the E field from the line array with transverse elements is

$$\overline{E}_q^{y,x'} = -\hat{y}\frac{Z}{\beta 4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z}$$
$$\cdot\, \frac{\partial}{\partial y}\int_{\text{Ref. ele.}} I'_{x'}(x')H_0^{(2)}(\beta r_\rho\sqrt{(x-x')^2+y^2})\, dx.' \tag{3.46}$$

The voltage induced in a y-directed element with current $I_y(y)$ is now given by (see Fig. 3.6, middle):

$$V_q^{y,x'} = \frac{1}{I_y(0)} \int_{\text{Ext. ele.}} I_y(y)\ E_q^{y,x'}\, dy. \tag{3.47}$$

Substituting (3.46) into (3.47), we obtain

$$\begin{aligned} V_q^{y,x'} = &-\frac{Z}{\beta 4 D_z} \sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} \frac{1}{I_y(0)} \int_{\text{Ref. ele.}} I'_{x'}(x')\, dx' \\ &\cdot \int_{\text{Ext. ele.}} I_y(y)\ \frac{\partial}{\partial y} H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2 + y^2})\, dy. \end{aligned} \tag{3.48}$$

Integrating the y integral once by parts, we get

$$\begin{aligned} V_q^{y,x'} = &-\frac{Z}{\beta 4 D_z} \sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} \frac{1}{I_y(0)} \int_{\text{Ref. ele.}} I'_{x'}(x')\, dx' \\ &\cdot \Bigg[I_y(y) H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2 + y^2})|_{\text{Ref. ele. endpts.}} \\ &- \int_{\text{Ext. ele.}} I'_y(y) H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2 + y^2}) dy \Bigg]. \end{aligned} \tag{3.49}$$

Since $I_y(y) = 0$ at the endpoints of the external element, the first term in (3.49) equals zero. Thus, (3.49) reduces to just the last term and for the mutual impedance between the line array and an external element we find

$$\begin{aligned} \overline{Z}_q^{y,x'} &= -\frac{\overline{V}_q^{y,x'}}{I_{x'}(0)} \\ &= -\frac{Z}{\beta 4 D_z} \sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} \frac{1}{I_{x'}(0) I_y(0)} \\ &\cdot \int_{\text{Ref. ele.}} I'_{x'}(x')\, dx' \int_{\text{Ext. ele.}} I'_y(y) H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2 + y^2})\, dy. \end{aligned} \tag{3.50}$$

For sinusoidal current distribution as given by (3.38) we find

$$\begin{aligned} I'_{x'}(x') &= \pm \beta I_a \cos\beta(l' - |x'|) \qquad \text{for} - l' < x' < l', \\ I_{x'}(0) &= I_a \sin\beta l'. \end{aligned}$$

Thus

$$Z_q^{y,z'} = \frac{Z}{4D_z \sin\beta l'} \sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} \frac{1}{I_y(0)} \int_{\text{Ref. ele.}} \text{sign } x \cos\beta(l' - |x'|)\, dx' \cdot \int_{\text{Ext. ele.}} I_y'(y) H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2 + y^2})\, dy. \tag{3.51}$$

Alternatively, we can simply in (3.44) carry out the differentiation $H_0^{(2)}$ with respect to x' and y as follows:

$$\frac{\partial}{\partial x'} H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2 + y^2}) = -\beta r_\rho H_0^{(2)'}(\beta r_\rho \sqrt{(x-x')^2 + y^2}) \frac{x-x'}{\sqrt{(x-x')^2 + y^2}}$$

$$\begin{aligned}
\frac{\partial}{\partial y}\left[\frac{\partial}{\partial x'} H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2 + y^2})\right]
&= -\beta r_\rho \left[\beta r_\rho H_0^{(2)''}(\beta r_\rho \sqrt{(x-x')^2 + y^2}) \cdot \frac{y}{\sqrt{(x-x')^2 + y^2}} \frac{x-x'}{\sqrt{(x-x')^2 + y^2}}\right. \\
&\quad \left. - H_0^{(2)'}(\beta r_\rho \sqrt{(x-x')^2 + y^2}) \frac{(x-x')y}{[(x-x')^2 + y^2]^{3/2}}\right] \\
&= -(\beta r_\rho)^2 \frac{(x-x')y}{(x-x')^2 + y^2}\left[H_0^{(2)''}(\beta r_\rho \sqrt{(x-x')^2 + y^2})\right. \\
&\quad \left. - \frac{1}{\beta r_\rho \sqrt{(x-x')^2 + y^2}} H_0^{(2)'}(\beta r_\rho \sqrt{(x-x')^2 + y^2})\right]
\end{aligned} \tag{3.52}$$

Use of the recursive formulas

$$H_n^{(2)'}(x) - \frac{n}{x} H_n^{(2)}(x) = -H_{n+1}^{(2)}(x)$$

yields

$$H_0^{(2)'} = -H_1^{(2)} \quad \text{and thus} \quad H_0^{(2)''} = -H_1^{(2)'}.$$

That is,

$$H_0^{(2)''} - \frac{1}{x} H_0^{(2)'} = -H_1^{(2)'} + \frac{1}{x} H_1^{(2)} = H_2^{(2)}. \tag{3.53}$$

Substituting (3.53) into (3.52), we get

$$\frac{\partial^2}{\partial x' \partial y} H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2 + y^2})$$

$$= -(\beta r_\rho)^2 \frac{(x-x')y}{(x-x')^2+y^2} H_2^{(2)}(\beta r_\rho \sqrt{(x-x')^2+y^2}). \tag{3.54}$$

Substituting (3.54) into (3.44), we obtain

$$\overline{E}_q^{y,x'} = -\hat{y}\frac{\beta Z}{4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} r_\rho^2$$
$$\cdot \int_{\text{Ref. ele.}} I_{x'}(x')\frac{(x-x')y}{(x-x')^2+y^2} H_2^{(2)}(\beta r_\rho \sqrt{(x-x')^2+y^2})\, dx.' \tag{3.55}$$

Substituting (3.55) into (3.47), we find for the voltage induced in a y-oriented element with current distribution $I_y(y)$:

$$V_q^{y,x'} = -\frac{\beta Z}{4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} r_\rho^2 \int_{\text{Ext. ele.}} \frac{I_y(y)}{I_y(0)}\, dy$$
$$\cdot \int_{\text{Ref. ele.}} I_{x'}(x')\frac{(x-x')y}{(x-x')^2+y^2} H_2^{(2)}(\beta r_\rho \sqrt{(x-x')^2+y^2})\, dx'. \tag{3.56}$$

Finally, for the mutual array impedance we have

$$Z_q^{y,x'} = -\frac{V_q^{y,x'}}{I_{x'}(0)}$$
$$= \frac{\beta Z}{4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} r_\rho^2 \int_{\text{Ext. ele.}} \frac{I_y(y)}{I_y(0)}\, dy$$
$$\cdot \int_{\text{Ref. ele.}} \frac{I_{x'}(x')}{I_{x'}(0)}\frac{(x-x')y}{(x-x')^2+y^2} H_2^{(2)}(\beta r_\rho \sqrt{(x-x')^2+y^2})\, dx'. \tag{3.57}$$

For further discussion see Section 3.5.

3.4.3 The z Component of $\overline{E}_q$

The z component $E_q^{z,x'}$ of the total field E_q from a line array with current distribution $I(x')$ is readily obtained from (3.31) by integration of the z component over the reference element:

$$\overline{E}_q^{z,x'} = \hat{z}\frac{j\beta}{\omega\varepsilon 4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} r_z$$
$$\int_{\text{Ref. ele.}} I'_{x'}(x')\frac{\partial}{\partial x'} H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2+y^2})\, dx'.$$

Noticing that $\dfrac{\partial}{\partial x} = -\dfrac{\partial}{\partial x'}$ and $\beta/\omega\varepsilon = Z$, we obtain

$$\overline{E}_q^{z,x'} = -\hat{z}\frac{jZ}{4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} r_z \times \int_{\text{Ref. ele.}} I_{x'}(x')\frac{\partial}{\partial x'} H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2+y^2})\, dx'. \tag{3.58}$$

Integrating by parts, we obtain

$$\overline{E}_q^{z,x'} = -\hat{z}\frac{jZ}{4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} r_z \left[I_{x'}(x') H_0^{(2)}\left(\beta r_\rho \sqrt{(x-x')^2+y^2}\right)\Big|_{\text{Ref. ele. endpt.}} - \int_{\text{Ref. ele.}} I'_{x'}(x') H_0^{(2)}\left(\beta r_\rho \sqrt{(x-x')^2+y^2}\right)\right] dx'. \tag{3.59}$$

As noted earlier, $I'_x(x') = 0$ at the endpoints of the reference elements [i.e., the first term in (3.59) is equal to zero], and we then obtain

$$\overline{E}_q^{z,x'} = \hat{z}\frac{jZ}{4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z-z')r_z} r_z \int_{\text{Ref. ele.}} I'_{x'}(x') H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2+y^2})\, dx'. \tag{3.60}$$

The voltage induced in a z-directed element with current $I_z(z)$ is now given by

$$V_q^{y,x'} = \frac{1}{I_z(0)}\int_{\text{Ext. ele.}} I_z(z)\; E_q^{z,x'}\, dz. \tag{3.61}$$

Substituting (3.60) into (3.61), we get

$$V_q^{z,x'} = \frac{jZ}{4D_z}\sum_{n=-\infty}^{\infty}\frac{r_z}{I_z(0)}\int_{\text{Ext. ele.}} I_z(z)\, e^{-j\beta(z-z')r_z}\, dz \cdot \int_{\text{Ref. ele.}} I'_{x'}(x') H_0^{(2)}(\beta r_\rho \sqrt{(x-x')^2+y^2})\, dx'. \tag{3.62}$$

The first integral in (3.62) is managed as usual by expressing z by a reference point $z^{(q)}$:

$$z = z^{(q)} + z'', \qquad -l_z < z < l_z. \tag{3.63}$$

Substituting (3.62) into (3.63) and noting that z' is a constant during the z integration, we obtain for the mutual impedance:

$$\begin{aligned} Z_q^{z,x'} &= -\frac{V^{z,x'}}{I_{x'}(0)} \\ &= -\frac{jZ}{4D_z}\sum_{n=-\infty}^{\infty} e^{-j\beta(z^{(q)}-z')r_z} r_z P_z \\ &\quad \cdot \frac{1}{I_{x'}(0)}\int_{\text{Ref. ele.}} I'_{x'}(x')H_0^{(2)}(\beta r_\rho\sqrt{(x-x')^2+y^2})\,dx', \end{aligned} \tag{3.64}$$

where the pattern function P_z is defined [similar to (3.24)] by

$$P_z = \frac{1}{I_{z''}(0)}\int_{\text{Ext. ele.}} I_z(z'')e^{-j\beta z'' r_z}\,dz''. \tag{3.65}$$

3.5 DISCUSSION

We recall from (3.4) that

$$r_\rho = \sqrt{1-r_z^2} = \sqrt{1-\left(s_z + n\frac{\lambda}{D_z}\right)^2}.$$

For the principal mode $n = 0$ and the incident field in real space (i.e., $|s_z| < 1$) we observe that r_ρ is always real. However, as $n \to \infty$ we also observe that r_ρ becomes imaginary, namely,

$$r_\rho = \underset{-}{(+)}\; j\sqrt{\left(s_z + n\frac{\lambda}{D_z}\right)^2 - 1} \tag{3.66}$$

[The choice of sign in (3.66) will soon become clear.]

For r_ρ imaginary the arguments in the Hankel functions become imaginary; that is, they are modified Hankel functions. Examples of the second kind of order 0 and 1, respectively, are shown in Fig. 3.7 for negative imaginary arguments. We observe a fast fall-off as the magnitude of the negative argument increases. In fact, the asymptotic value is given by

$$H_v^{(2)}(x) \underset{x\to\infty}{\longrightarrow} \sqrt{\frac{2j}{\pi x}}\, j^v\, e^{-jx}, \tag{3.67}$$

that is, it drops off somewhat faster than an exponential term when x is negative imaginary.

Thus, the infinite series (3.27) will in spite of the term $r_\rho^2 \sim (s_z + n\lambda/D_z)^2$ for n large converge very fast. In fact, in Problem 3.3 you are asked to compare the convergence of the impedance of an infinitely long column array given by (3.27)

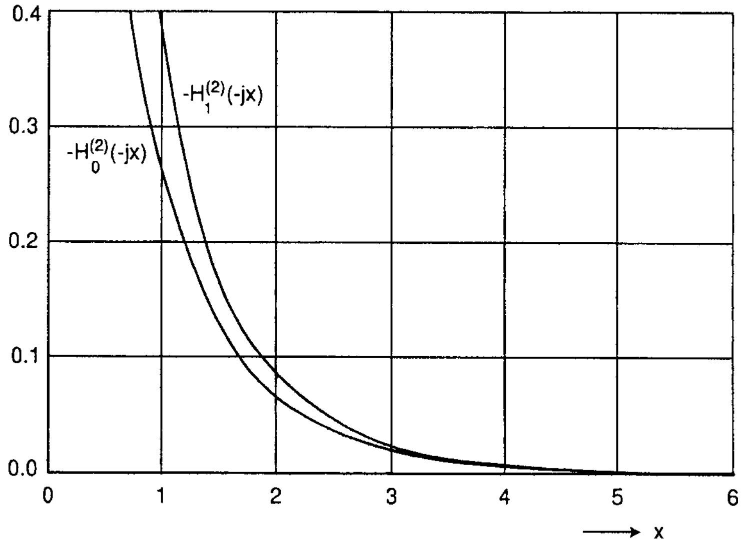

Fig. 3.7 The Hankel function of the second kind and orders 0 and 1, respectively, for negative imaginary argument. For asymptotic approximation, see (3.67).

with the impedance of an infinite × infinite array given by (3.28). The latter is of course known to converge very fast as observed in the PMM code.

Let us summarize our investigation so far.

In the *longitudinal* case we found that both the z and ρ components of the field could be expressed by a single fast convergent series of Hankel functions [see (3.16)]. Furthermore, the mutual impedance between the collinear array and external element parallel with the array elements (i.e., z-directed in our case) could also be expressed by a simple fast convergent series [see (3.27)]. However, if the external element was ρ-directed, simple integration was not possible.

In the *transverse* case we found that the field parallel with the array elements (i.e., x-directed in our case) could be expressed by a simple fast convergent infinite series containing three Hankel functions [see (3.39)]. However, to find the mutual impedance as given by (3.43), numerical integration will be necessary.

Furthermore, the field components orthogonal to the x-directed elements, namely the y- and z-directed components as expressed for example by (3.55) and (3.60), respectively, has so far not been possible to integrate into a simpler form.

It was primarily for that reason that Usoff decided to work in the spatial domain when he wrote the SPLAT program. He further used a shanks transform to obtain faster convergence [24]. It became a most wonderful and versatile program. In fact, it is the workhorse for our research into finite arrays.

However, it was always the author's dream to work in the spectral domain similar to the PMM program that is very fast converging and gives us the real part of the impedance as one term if no grating lobes are present. Thus, if anyone ever pulls it off, the author would appreciate to hear about it.

3.6 DETERMINATION OF THE ELEMENT CURRENTS

When all the mutual and the self-impedances of an assembly of stick arrays as shown in Fig. 3.8 has been determined, it is a relatively simple matter to find the currents on the elements. In fact, it was already discussed in my earlier book [62] and need therefore only to be restated here for easy reference.

Let the stick arrays in Fig. 3.8 be exposed to an incident plane wave with direction of propagation equal to $\hat{s}$. The voltages induced in the reference element of the three or more stick arrays are denoted $V^{(1)}$, $V^{(2)}$, and $V^{(3)}$, respectively. They are easily determined by application of (3.18). Similarly, the currents on each reference element are denoted by $I^{(1)}$, $I^{(2)}$, and $I^{(3)}$, respectively.

Then from the generalized Ohm's Law we obtain the following matrix equation:

$$\begin{bmatrix} V^{(1)} \\ V^{(2)} \\ V^{(3)} \end{bmatrix} = \begin{bmatrix} [Z^{1,1} + Z_{L1}] & Z^{1,2} & Z^{1,3} \\ Z^{2,1} & [Z^{2,2} + Z_{L2}] & Z^{2,3} \\ Z^{3,1} & Z^{3,2} & [Z^{3,3} + Z_{L3}] \end{bmatrix} \begin{bmatrix} I^{(1)} \\ I^{(2)} \\ I^{(3)} \end{bmatrix}. \quad (3.68)$$

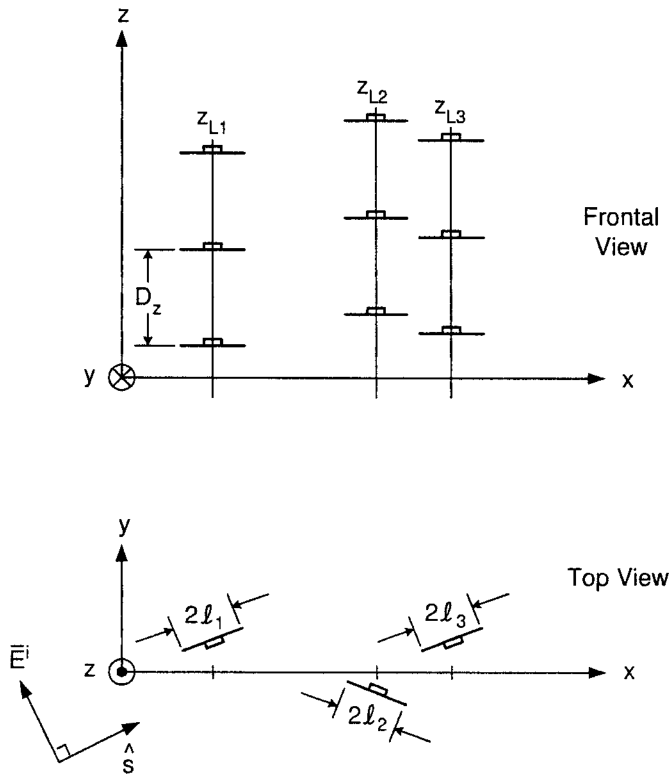

Fig. 3.8 An arbitrary but finite number of infinitely long stick arrays with the same interelement spacing D_z is exposed to an incident plane wave $\overline{E}_i$. Element orientation and length may vary from stick array to stick array.

Here $Z^{k,m}$ denotes the mutual array impedances between the reference element in array k and all the elements in array m while Z_{L1}, Z_{L2}, and Z_{L3} denote the load impedances in the respective array elements.

Note that the element orientation can vary from stick to stick while the interstick spacing can be arbitrary. Only the element spacing D_z in the z direction must be the same from stick to stick. Otherwise, Floquet's Theorem will be violated.

3.7 THE DOUBLE INFINITE ARRAYS WITH ARBITRARY ELEMENT ORIENTATION

In my first book we found the vector potential as well as the magnetic and electric fields for double infinite arrays of arbitrarily oriented Hertzian elements of length dl [63]. However, our investigation of elements of arbitrary length was basically limited to elements contained in the plane of the array (but of arbitrary shape and orientation in that plane).

It was also pointed out that several of the author's former students and others have tackled that problem [66–71]. The reason for not considering it further in my first book was that it leads to a very messy nested integration that could lead to certain convergence problems. The following derivation by Peter Munk avoids this problem, and we therefore find it appropriate to present it here.[2]

3.7.1 How to Get Well-Behaved Expressions

Consider an infinite array of elements oriented in the $\hat{p}^{(1)}$ direction located in slab m of a stratified dielectric medium as shown in Fig. 3.9. As throughout [61], the elements are assumed to be excited by an external plane wave. The plane wave spectrum generated by this array can be decomposed into five wave modes as described in references 66–71. Four of these modes involve the reflection coefficient from the dielectric interfaces bounding the m_{th} slab, while the fifth mode does not and is referred to as the "direct mode." It is this mode that requires special treatment and is the reason it is revisited in detail here. The direct mode field at some point $\overline{R}$ on a test element also located entirely in slab m and oriented in the $\hat{p}$ direction may be written [66, 68]

$$\overline{E}(\overline{R}) = \frac{Z_m}{2D_x D_z} \sum_{k=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} \left\{ \frac{e^{-j\beta_m(\overline{R}-\overline{R}^{(1)})\cdot\hat{r}_{m+}}}{r_{my}} \right.$$

[2] While a graduate student at OSU, Peter took all the courses offered by the author. He was never a student of his since that would have been inappropriate. However, unofficially a good deal of consultation took place. Someone once stated that a good teacher at some point starts learning from his students. While the author never made any claims of being a particularly good teacher, he can at least testify to the fact that he learned a lot from all his students.

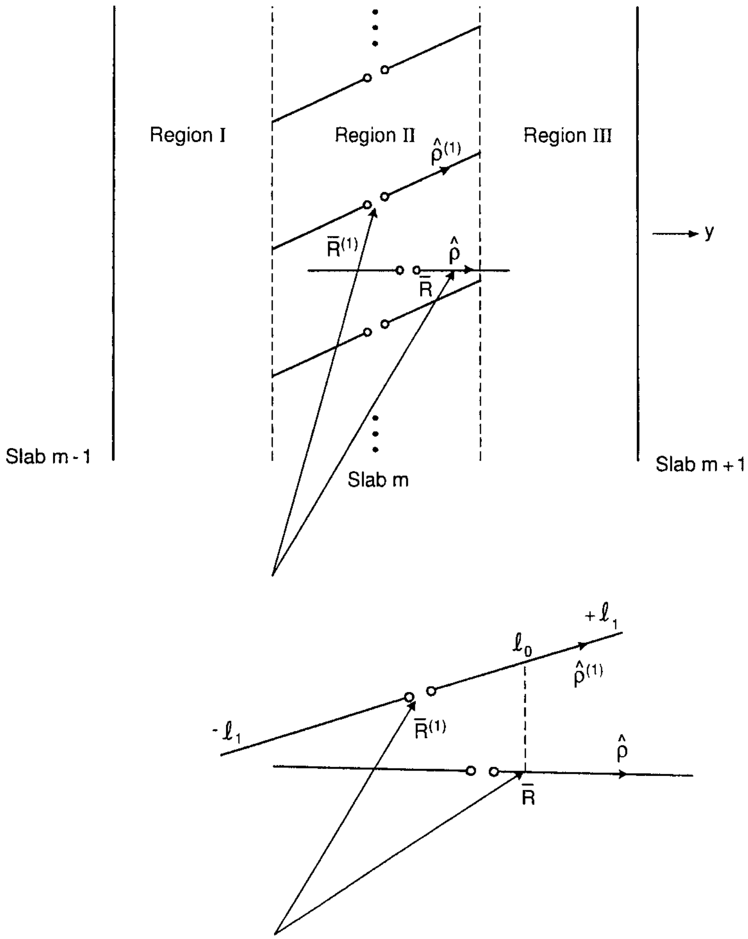

Fig. 3.9 Top: An array of arbitrarily oriented dipoles located in the dielectric slab m sandwiched between slab m − 1 and m + 1. Regions I, II, and III are defined by the tips of the dipoles as shown in the figure. Bottom: Detail of the element geometry.

$$\cdot\, \overline{e}_{+} U(l_1 + l_o) \int_{-l_1}^{l_o} I_o^{(1)}(l)\, e^{j\beta_m l \hat{p}^{(1)}\cdot\hat{r}_{m+}}\, dl$$

$$+ \frac{e^{-j\beta_m(\overline{R}-\overline{R}^{(1)})\cdot\hat{r}_{m-}}}{r_{my}} \cdot \overline{e}_{-} U(l_1 - l_o) \cdot \int_{l_o}^{l_1} I_o^{(1)}(l)\ e^{j\beta_m l \hat{p}^{(1)}\cdot\hat{r}_{m-}} dl \Bigg\}, \quad (3.69)$$

where the Heavyside unit step functions $U(l_1 - l_o)$ and $U(l_1 + l_o)$ illustrate that if $\overline{R}$ is located in either region I or III, only the $\hat{r}_{m-}$ or $\hat{r}_{m+}$ waves are observed, respectively. In addition, l_o is defined as the y projection of the field point $\overline{R}$

onto the reference element and may be expressed as [66,68]

$$l_o = \frac{1}{p_y^{(1)}}(R_y - R_y^{(1)}), \tag{3.70}$$

where

$$\overline{R} \equiv R_x\hat{x} + R_y\hat{y} + R_z\hat{z}, \tag{3.71}$$

and the point $\overline{R}'$ along the reference element is given by

$$\overline{R}' = \overline{R}^{(1)} + l\hat{p}^{(1)}, \tag{3.72}$$

for $-l_1 \leq l \leq +l_1$, where

$$\overline{R}^{(1)} \equiv R_x^{(1)}\hat{x} + R_y^{(1)}\hat{y} + R_z^{(1)}\hat{z}. \tag{3.73}$$

Examining the first integral in (3.69) and employing integration by parts, one may write

$$\int_{-l_1}^{l_o} I_o^{(1)}(l)\, e^{j\beta_m l\hat{p}^{(1)}\cdot\hat{r}_{m+}} dl = \frac{I_o^{(1)}}{j\omega_+} e^{j\omega_+ l}\bigg|_{-l_1}^{l_o} - \frac{1}{j\omega_+}\int_{-l_1}^{l_o} \frac{\partial I_o^{(1)}(l)}{\partial l} e^{j\omega_+ l}\, dl, \tag{3.74}$$

where $\omega_\pm = \beta_m\hat{p}^{(1)}\cdot\hat{r}_{m\pm}$. The second integral may be treated similarly, yielding the expression

$$\int_{l_o}^{l_1} I_o^{(1)}(l) e^{j\beta_m l\hat{p}^{(1)}\cdot\hat{r}_{m-}} dl = \frac{I_o^{(1)}}{j\omega_-} e^{j\omega_- l}\bigg|_{l_o}^{l_1} - \frac{1}{j\omega_-}\int_{l_o}^{l_1} \frac{\partial I_o^{(1)}(l)}{\partial l} e^{j\omega_- l}\, dl. \tag{3.75}$$

Both integrals in (3.74) and (3.75) are well-behaved and will converge rapidly inside the double infinite summation, as will the terms evaluated at $+l_1$ and $-l_1$. However, these same two terms evaluated at l_o will exhibit convergence problems and must therefore be treated separately. It is beneficial at this point to decompose these two terms into their respective *x, y*, and *z* components. Making use of the fact that $\overline{e}_\pm$ may be written [64]

$$\overline{e}_\pm = \hat{x}\left(\frac{\omega_\pm r_{mx}}{\beta_m} - p_x^{(1)}\right) + \hat{y}\left(\frac{\pm\omega_\pm r_{my}}{\beta_m} - p_y^{(1)}\right) + \hat{z}\left(\frac{\omega_\pm r_{mz}}{\beta_m} - p_z^{(1)}\right), \tag{3.76}$$

the $\hat{x}$ components of the two terms may be combined, resulting in

$$\frac{I_o^{(1)}(l_o)\Lambda_o p_x^{(1)}}{j\omega_- r_{my}} - \frac{I_o^{(1)}(l_o)\Lambda_o p_x^{(1)}}{j\omega_+ r_{my}}, \tag{3.77}$$

where

$$\Lambda_o \equiv e^{-j\beta_m[(R_x - R_x^{(1)} - p_x^{(1)} l_o) r_{mx} + (R_z - R_z^{(1)} - p_z^{(1)} l_o) r_{mz}]}. \tag{3.78}$$

Likewise, combining the $\hat{z}$ components yields

$$\frac{I_o^{(1)}(l_o)\Lambda_o p_z^{(1)}}{j\omega_- r_{my}} - \frac{I_o^{(1)}(l_o)\Lambda_o p_z^{(1)}}{j\omega_+ r_{my}}. \tag{3.79}$$

Examining (3.77) and (3.79) reveals that both expressions converge nicely in the double infinite summation over k and n. Combining the $\hat{y}$ components, on the other hand, results in an expression that differs in form from the combined $\hat{x}$ and $\hat{z}$ components and may be expressed instead as

$$-j\Lambda_o I_o^{(1)}(l_o) \cdot \left\{ \frac{2}{\beta_m} + \frac{p_y^{(1)}}{\omega_-} - \frac{p_y^{(1)}}{\omega_+} \right\}. \tag{3.80}$$

The second and third terms in (3.80) are identical in form to those found in (3.77) and (3.79), and likewise they converge rapidly in the double infinite summation. The convergent nature of the first term, however, is not immediately apparent. Employing the same mathematical identity used in references 66 and 68, the double infinite summation of the first term in (3.80) involving Λ_o may be written

$$\begin{aligned} &\frac{-j2I_o^{(1)}(l_o)}{\beta_m} \cdot \sum_{k=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} \Lambda_o \\ &= \frac{-j2I_o^{(1)}(l_o)}{\beta_m} \cdot e^{-j\beta_m [s_{mx}(R_x - R_x^{(1)} - p_x^{(1)} l_o) + s_{mz}(R_z - R_z^{(1)} - p_z^{(1)} l_o)]} \\ &\cdot D_x D_z \cdot \sum_{k=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} \delta[(R_x - R_x^{(1)} - p_x^{(1)} l_o) - kD_x] \\ &\cdot \delta[(R_z - R_z^{(1)} - p_z^{(1)} l_o) - nD_z]. \end{aligned} \tag{3.81}$$

The dirac delta functions in (3.81) are identically zero unless

$$R_x - R_x^{(1)} - p_x^{(1)} \cdot l_o = kD_x, \tag{3.82}$$

for $k = 0, \pm 1, \pm 2, \ldots$, and

$$R_z - R_z^{(1)} - p_z^{(1)} \cdot l_o = nD_z, \tag{3.83}$$

for $n = 0, \pm 1, \pm 2, \ldots$.

Considering only the cases where the Heavyside unit step functions in (3.69) differ from zero (i.e., the nontrivial case where $\overline{r}$ overlaps in the $\hat{y}$ direction with the reference element), it is observed that (3.82) and (3.83) can never be true unless the test element is in contact with one (or more) of the reference array elements. Since this is not the case, the first term in (3.80) may be dropped.

Recombining the individual *x, y*, and *z* components then yields this new, "well-behaved" *E*-field equation

$$\overline{E}(\overline{R}) = \frac{jZ_m}{2D_xD_z}\sum_{k=-\infty}^{+\infty}\sum_{n=-\infty}^{+\infty}\left\{\frac{e^{-j\beta_m(\overline{R}-\overline{R}^{(1)})\cdot\hat{r}_{m+}}}{r_{my}\omega_+}\left[I_o^{(1)}(-l_1)e^{-j\omega_+ l_1}\right.\right.$$
$$\left. + \int_{-l_1}^{l_o}\frac{\partial I_o^{(1)}(l)}{\partial l}e^{j\omega_+ l}dl\right]\overline{e}_+ + \frac{I_o^{(1)}(l_o)\Lambda_o}{r_{my}\omega_+}\hat{p}^{(1)}$$
$$- \frac{e^{-j\beta_m(\overline{R}-\overline{R}^{(1)})\cdot\hat{r}_{m-}}}{r_{my}\omega_-}\left[I_o^{(1)}(l_1)e^{j\omega_- l_1} - \int_{l_o}^{l_1}\frac{\partial I_o^{(1)}(l)}{\partial l}e^{j\omega_- l}dl\right]\overline{e}_-$$
$$\left. - \frac{I_o^{(1)}(l_o)\Lambda_o}{r_{my}\omega_-}\hat{p}^{(1)}\right\}. \tag{3.84}$$

Thus, the *E* field written in the form of (3.84) does not have the convergence problems observed earlier and should consequently be used in the architecture for a future PMM program with arbitrarily oriented nonplanar elements.

3.8 CONCLUSIONS

In this chapter we have obtained the *E* field from stick arrays composed of longitudinal or transverse elements of arbitrary length $2l$.

In the *longitudinal* case we found that the field could always be expressed by a single infinite series of Hankel functions multiplied with a simple pattern functions [see (3.17)]. Similarly, the mutual impedance between the array and an external element parallel to the longitudinal elements could be obtained by further multiplication of a pattern function associated with the external element [see (3.27)]. If the external element was orthogonal to the longitudinal elements, the mutual impedance could be obtained by numerical integration of the orthogonal fields along the external elements.

In the *transverse* case we found that the electric fields parallel to the elements could be expressed by a single infinite series as the sum of just three Hankel functions emanating from the endpoints and the center of the elements, respectively. The mutual impedance to an external element could then be obtained by numerical integration of these three Hankel functions.

For the mutual impedance between an array of transverse elements and an external orthogonal element we found that a numerical double integration was necessary.

The computational advantage of using the Hankel functions is that they contain the argument $\beta r_\rho \rho$. Thus, as we enter the imaginary space, r_ρ would become imaginary (see Section 3.4), leading to modified Hankel functions that fall off exponentially as ρ or n increase (see Fig. 3.7). Thus, we could obtain the mutual impedances between arrays with elements of arbitrary orientation. Once these were determined, it was a relatively simple matter to obtain the element currents (see Section 3.6).

Most of these expressions were not available to Usoff when he was given the task to write a program for finite arrays. He therefore chose to stay mostly in the spatial domain except for the longitudinal case. The result was a program that was extremely versatile and became the workhorse for our finite array research. It was later given a "face lift" by Dan Janning and other students that reduced the run time and made it more user friendly.

It is doubtful whether any of the author's students will ever attempt to write a program in the spectral domain for finite arrays like originally envisioned (I am running out of time, students, and money!). But if anyone out there does, please let me hear about it.

PROBLEMS

3.1 Consider a stick array with longitudinal elements as shown in Fig. 3.2 with element lengths $2l = 1.5$ cm and $D_z = 1.6$ cm. The incident field is arriving at broadside with $\hat{s} = \hat{y}$ and $\overline{E}^i = \hat{z}E^i$. Furthermore, the wire radius a is 0.1 cm. Assume sinusoidal current distribution.

1. At $f = 10$ GHz evaluate the terms $n = 0, \pm 1, \pm 2$, and ± 3 in (3.26).
2. How many of these terms contain a real part?
3. At what value of D_z do we encounter the first grating lobe?
4. If D_z is so small that no grating lobes are encountered, how does the radiation resistance R_A vary with D_z (constant frequency).
5. Does the radiation resistance R_A vary significantly with the wire radius "a," if at all?
6. Does the $n = 0$ term depend on the wire radius "a"?

3.2 An infinite × infinite array has the elements oriented along $\hat{p} = \hat{z}$ with interelement spacings $D_x = 0.8$ cm and $D_z = 1.6$ cm. The total length of the elements are $2l = 1.5$ cm and the wire radius $a = 0.1$ cm. An electromagnetic wave is incident upon this array at broadside; that is, $\hat{s} = \hat{y}$, while $\overline{E}^i = \hat{z}E^i$. Assume sinusoidal current distribution.

1. For $f = 10$ GHz, evaluate the terms in (3.28) for $n = 0, \pm 1, \pm 2, \pm 3$ and $k = 0$.
2. How many of these terms are real?
3. If we assume that D_x and D_z are so small that no grating lobes exist, how does the radiation resistance R_A vary with these interelement spacings?
4. Does R_A depend strongly on the wire radius "a," if at all?

3.3 Compare the convergence properties for the stick array in Problem 3.1 with that of the infinite × infinite array in Problem 3.2.

3.4 A stick array with transverse elements as shown in Fig. 3.4 has the interelement spacing $D_z = 0.8$ cm and the reference element located at the origin. The total length of the elements are $2l = 1.5$ cm, and the wire radius a is 0.1 cm. An EM field is incident at broadside with $\hat{s} = \hat{y}$ and $\overline{E}^i = \hat{x}E^i$.

Evaluate the terms $n = 0, \pm 1, \pm 2$ and ± 3 in (3.39) when the point of observation is located at

1. $\overline{R} = (0, 1, 0)$ cm
2. $\overline{R} = (\pm 1.5, 1, 0)$ cm

3.5 The E field for an infinite × infinite array with $\hat{p} = \hat{x}$ is given in reference 63 by

$$\overline{E}(\overline{R}) = I(\overline{R}^{(1)}) \frac{Z}{2D_x D_z} \sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{e^{-j\beta(\overline{R}-\overline{R}^{(1)})}}{r_y} \overline{e}_{\pm} P,$$

where

$$\overline{e}_{\pm} = (\hat{p} \times \hat{r}_{\pm}) \times \hat{r}_{\pm},$$

$$P = \frac{1}{I(\overline{R}^{(1)})} \int_{\text{Ref. Ele.}} I(l)\, e^{j\beta l \hat{p}\cdot\hat{r}_{\pm}}\, dl,$$

$$r_y = \sqrt{1 - \left(s_x + k\frac{\lambda}{D_x}\right)^2 - \left(s_z + n\frac{\lambda}{D_z}\right)^2}.$$

An E field is incident at broadside with $\overline{E}^i = \hat{x}E^i$, $D_x = 1.6$ cm, and $D_z = 0.8$ cm. The total element length $2l$ is 1.5 cm, and the wire radius a is 0.1 cm.

Compare the convergence properties of the E field for the stick array in Problem 3.4 with that of the infinite × infinite array in this problem.